University of Eloued

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Destination to: Master GRH and Management

Classic regression Vs trained Regression

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1.First Example: Classic regression Vs trained Regression

This section contrasts the traditional statistical approach to regression with the modern machine learning paradigm. Classical regression emphasizes inference, utilizing the entire dataset to estimate coefficients, evaluate statistical significance (such as p-values), and interpret the underlying relationships between variables. In contrast, the "trained" machine learning approach prioritizes predictive generalization, employing data splitting techniques—specifically dividing data into training and testing sets—to rigorously evaluate how well the model performs on unseen data. The following practical example applies both methodologies to the same dataset, highlighting the fundamental shift from explaining data in-sample to predicting outcomes out-of-sample.

1.Linear Regression

Let's take the next dataset:

Depended variable: Exam_Score **Linear Regression**

Hours_Studied	Exam_Score
1	5
2	6
3	4
4	8
5	7
6	9
7	7
8	5

Logistic Regression

Hours_Studied	Exam_Score
1	0
2	0
3	1
4	0
5	1
6	0
7	1
8	1

Questions:

Part _One: Simple dataset (concerned both of Linear and the Logistic dataset)

Part 1: Classical Estimation (In-Sample Approach)

1. Using the statsmodels library, write the Python code to estimate a classical Ordinary Least Squares (OLS) regression model for Exam_Score based on Hours_Studied using the **entire dataset** (8 observations). Ensure you include the necessary line of code to add a constant (intercept) and the command to display the statistical summary.

2. Based on the classical model fitted to the full dataset, write the Python code required to calculate the in-sample Mean Squared Error (MSE). Furthermore, explain conceptually why this in-sample metric is insufficient for evaluating the model's true ability to predict unseen data.

Part 2: Manual Data Partitioning (Data Splitting)

3. To transition to a predictive machine learning methodology (Trained Model), we will manually split our small dataset without relying on built-in functions like train_test_split. Write the Python code using standard array indexing or list slicing to partition the variables X and y. Assign the first 6 observations to the training set (X_train, y_train) and the final 2 observations to the testing set (X_test, y_test).

Part 3: Model Training, Prediction, and Evaluation Metrics

4. Utilizing the scikit-learn library, write the programmatic steps to instantiate a LinearRegression model, train (fit) it **exclusively** on the training data (X_train, y_train), and generate predictions (y_pred) for the testing data (X_test).

5. Given the predictions generated for the test set, write the required Python code to calculate both the Mean Squared Error (MSE) and the Root Mean Squared Error (RMSE) using the appropriate functions from the sklearn.metrics module.

6.

Part 4: Methodological Comparison (Classic vs. Trained Paradigms)

Analytical Question: Compare the in-sample MSE obtained from the classical model (fitted on the full dataset) with the out-of-sample test MSE obtained from the trained machine learning model. If the out-of-sample error is inherently higher, how do you mathematically and logically explain this phenomenon through the lens of model "generalization"? Discuss the fundamental difference between explaining existing historical data and forecasting future, unseen outcomes.

Part _Two: Multiple dataset

Part 1: Classical Multiple Regression (In-Sample Estimation)

1. Using the statsmodels library, write the Python code to estimate a classical Ordinary Least Squares (OLS) multiple regression model predicting Exam_Score using a feature matrix X that contains three independent variables: Hours_Studied, Attendance_Rate, and Previous_GPA. Explicitly include the code to add a constant term to the matrix and display the comprehensive statistical summary.

2. In the resulting classical statistical summary, you will observe both R^2 and Adjusted R^2. Explain the mathematical and logical necessity of relying on Adjusted R^2 rather than the standard R^2 when evaluating an in-sample multiple regression model with numerous predictors.

Part 2: Automated Data Partitioning (Data Splitting)

3. Moving beyond manual slicing for multidimensional arrays, write the Python code utilizing sklearn.model_selection.train_test_split to divide your feature matrix X and target vector y into training (80%) and testing (20%) sets. Explain the programmatic and pedagogical importance of setting the random_state parameter when splitting a dataset for machine learning.

Part 3: Model Training, Prediction, and Feature Extraction

4. Utilizing the scikit-learn library, write the complete Python workflow to instantiate a LinearRegression model, train (fit) it exclusively on the multidimensional X_train and y_train arrays, and generate a vector of predictions (y_pred) for the unseen X_test data.

5. In multiple regression, the model generates multiple partial slope coefficients ($\beta_1, \beta_2, \beta_3$). Write the scikit-learn code required to extract and print the model's intercept and its array of trained coefficients. Conceptually, how

does the machine learning reliance on these coefficients differ from the classical statistical approach (which relies heavily on coefficient p-values and confidence intervals)?

Part 4: Methodological Comparison (Overfitting and Complexity)

6. **Analytical Question:** A student adds five new, highly correlated, and mostly irrelevant variables to the dataset. In the classical statsmodels approach, the standard in-sample R^2 artificially increases. However, in the scikit-learn predictive approach, the out-of-sample Test MSE drastically worsens. Explain this exact phenomenon by contrasting the classical concept of "Multicollinearity" with the machine learning concepts of "Overfitting" and the "Bias-Variance Tradeoff."

Solution

you can find the full solution under the below next Links:

1.Linear regression (Simple and Multiple):

https://colab.research.google.com/drive/130E231RtXAzO3JAL7YeCXA__YEeok5PF#scrollTo=c_5farHBerHB

2.Logistic Regression (Simple and Multiple):

https://github.com/atman500/Deep_Learning/blob/main/copy_of_trained_Logistic_Vs_Classic_models.ipynb

or:

https://colab.research.google.com/drive/1EhMmMFTMISYkffwufyOjkTpSPIS_X6wS#scrollTo=AW2q8mlWSEKb